Fundamentals of Programming: Final Examination

Remarks

- There are four programs in this final exam.
- You can **ONLY** refer to the **lecture materials**.
- **Never use generative AI** and **Never communicate/share answers with others. Turn off Copilot** if it is installed in your VS Code.
- The time limit is **100 minutes**. You can leave the room whenever you have finished.
- Please name your programs exactly as specified: `bmi.c` , `area.c` , `coffee.c` , and `election.c` . If you skip a program, you do not have to submit that file.
- **Strictly follow the input/output format.** Do not add extra conversational text like `Please input:` or `The answer is:` as it will cause the evaluation to be more complex.

1. BMI Category (`bmi.c`)

You will receive a person's BMI (Body Mass Index) as a floating-point number. Your task is to output their weight category based on the exact table provided in your lecture materials:

BMI	Result
< 18.5	Underweight
18.5 - 25	Normal
25 - 30	Overweight
> 30	Obese

Input

You will receive the input on a single line:

```
bmi
```

Where `bmi` is a floating-point number.

Output

Print the exact category name (`Underweight` , `Normal` , `Overweight` , or `Obese`) on a single line.

Sample 1

Input:

```
22.5
```

Output:

```
Normal
```

Sample 2

Input:

```
31.2
```

Output:

```
Obese
```

Scoring

This problem is worth 20 points.

2. Geometric Area Engine (`area.c`)

You are writing a simple area calculation engine. You will receive a single character denoting a shape type, followed by two integer parameters.

The shape character will be one of the following:

- `R` (Rectangle): The parameters are width and height. Output the area.
- `T` (Triangle): The parameters are base and height. Output the integer area (assume base $\times$ height is always an even number for this test).
- `S` (Square): The first parameter is the side length. The second parameter will be given but should be **ignored**. Output the area.

Input

You will receive the input in the following format:

```
Shape param1 param2
```

Where `Shape` is `R` (Rectangle), `T` (Triangle), or `S` (Square), and the parameters are integers between 1 and 1000.

Output

Please output the calculation result on a single line.
Any other explanations or strings are strictly forbidden.

Sample 1

Input:

```
R 10 5
```

Output:

```
50
```

Sample 2

Input:

```
S 8 99
```

Output:

```
64
```

Scoring

This program is worth 25 points.

3. Coffee Roaster Queue (`coffee.c`)

You are managing the automated queue for a boutique coffee roaster.
You are given a string of n consecutive customer orders. Each order is either for a Light Roast (`L`) or a Dark Roast (`D`).

The roaster operates on the following rules:

- The machine tracks pending orders for both Light and Dark roasts starting at $0:0$.
- When a customer orders, their roast type gets one point in the queue.
- **The machine can only roast batches of exactly 4.** As soon as a roast type reaches 4 pending orders, the machine fires, printing the type it roasted (`L` or `D`), and the queue for **both** types is completely cleared and reset to $0:0$ (the unroasted beans of the other type are discarded for freshness).

Your task is to print the sequence of roasts that occur, and the final leftover queue state.

Input/Output Format

The input is a string of length n, consisting only of the characters `L` and `D`. The number of orders n is not explicitly given; you must determine it from the input string (the maximum length is 100,000).

The output should be a list, with each line representing a completed roast (`L` or `D`). If there are leftover pending orders at the end of the string, output them on the final line in the format `L:D`.

Sample 1

Input:

```
LLLDDLLLDDL
```

Output:

```
L
D
L:0
```

(Explanation: 3 Ls, then 2 Ds, then the 4th L arrives. The machine roasts 'L' and resets. Then 2 Ls and 2 Ds arrive, making 4 total Ds. The machine roasts 'D' and resets. The final 'L' is left in the queue,

printing 'L:0').

Scoring

This problem is worth 25 points.

4. Election Polling (`election.c`)

You will receive an $r \times c$ matrix representing the votes cast across r districts for c political candidates. Your task is to calculate the total votes per district (rows) and the total votes per candidate (columns).

Given an $r \times c$ matrix (e.g., $r = 3$ districts and $c = 4$ candidates):

```
1 4 3 7
3 4 3 6
9 9 3 5
```

Calculate the summation along the row and put the result D_i as the $(c+1)$-th element.
Then, calculate the summation along the column and put the result C_i as the $(r+1)$-th element.
The very bottom-right element should be the grand total of all votes.

Input

You will receive the row size r and column size c on the first line.
The following r lines contain the $r \times c$ matrix.

```
r c
e_11 e_12 ... e_1c
e_21 e_22 ... e_2c
.
.
e_r1 e_r2 ... e_rc
```

$r, c \leq 100$ and all inputs are positive integers.

Output

Please output the $(r+1) \times (c+1)$ matrix.

```
e_11 e_12 ... e_1c D_1
e_21 e_22 ... e_2c D_2
.
.
C_1  C_2  ... C_c  TOTAL
```

Sample 1

Input:

```
2 3
5 5 2
10 2 1
```

Output:

```
5 5 2 12
10 2 1 13
15 7 3 25
```

Scoring

This problem is worth 30 points.